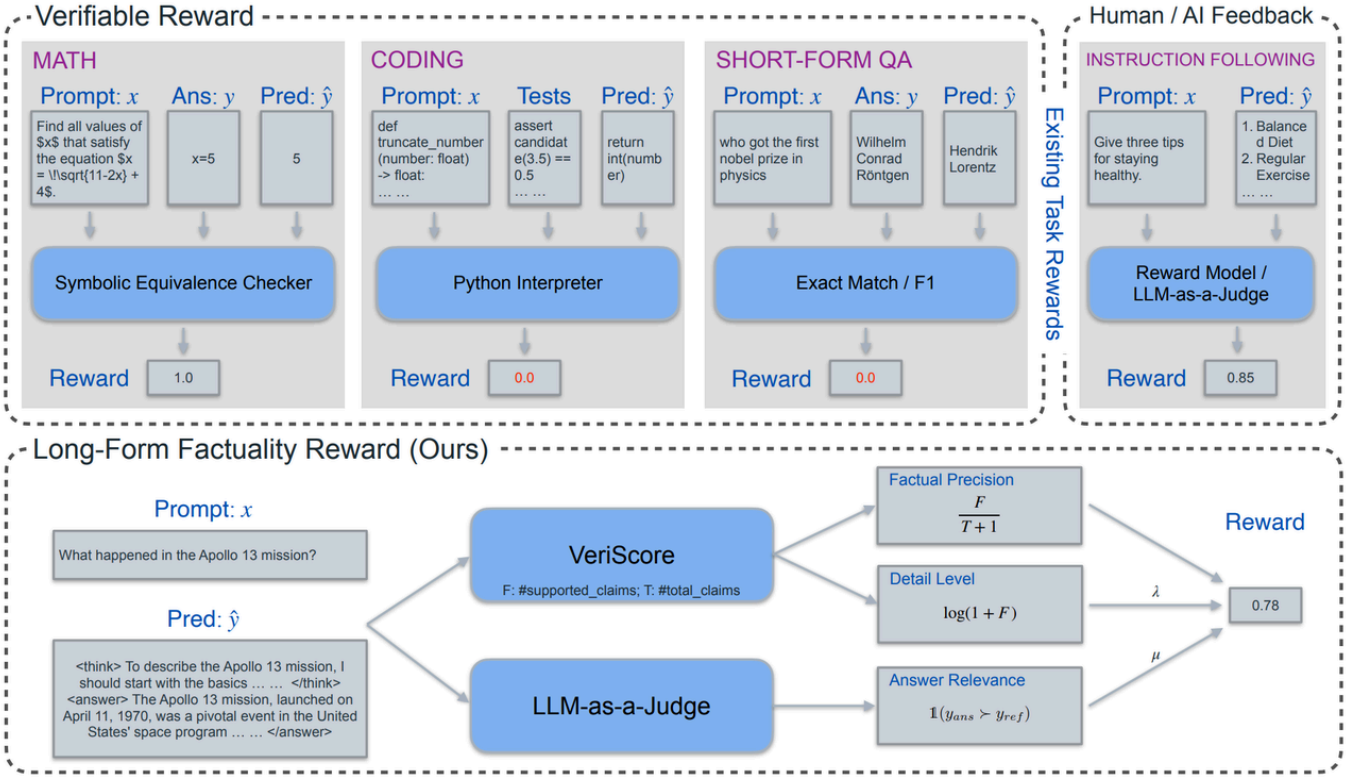
Learning to Reason for Factuality

> #RL #Finetuning #CoT #GRPO



- What problems does this framework solve?
  - It is observed that thinking models, when handling some very complex tasks, will even have more hallucination
  - It is found that their hallucination rates are, on average, **10 and 13 percentage points higher** than those of DeepSeek-V3 and Qwen-2.5-32B, respectively.
- What does this framework do?
  - This paper proposes a compound reward to ensure the usage of CoT, the quality of answer, the factuality, and the diversity of factuality. For each output y of each input x:

$$\mathcal{R}(y|x) = \begin{cases} -1.0, & \text{if } y \text{ is malformed} \\ \mathcal{R}_{fact} + \lambda \cdot \mathcal{R}_{dtl} + \mu \cdot \mathcal{R}_{rel} \quad = \quad \frac{F}{T+1} + \lambda \cdot \log(1+F) + \mu \cdot \mathbb{1}(y_{ans} \succ y_{ref}), & \text{otherwise} \end{cases}$$

    - If the answer is *malformed* (i.e., not following CoT token `<think>...</think>`), then the reward is -1.
    - $\mathcal{R}_{fact}$: the proportion of verified facts over total claims ($\frac{F}{T+1}$). The +1 term on the denominator is to void a denominator of zero.
    - $\mathcal{R}_{dtl}$: the number of verified facts. The author used a logarithmic function instead of using F directly. This was done to account for diminishing marginal returns.
    - $\mathcal{R}_{rel}$: check whether the generated answer is better than the reference answer.
  - To verify the factuality of answers, **VeriScore** is used. It will extract the claims mentioned in the response and verify the claims. It is modified to increase the efficiency by batching, asynchronization, and parallel processes.
  - Steps
    - Data Curation: use a model to generate a set of prompts X.

        - SFT: teach the model to follow the `<think>...<\think>`.
        - For each prompt $x \in X$, generate N answers.
        - Calculate the compound reward.
        - Run RL (GRPO).
- Key Findings
    1. Hallucinations are significantly reduced: Compared to the base model, this method reduces hallucination rates by 23.1 percentage points (i.e., substantially improving factual accuracy).
    2. Increased detail: The number of correct facts included in responses increased by 23%, demonstrating that the model does not cheat by becoming "silent."
    3. Quality maintained: The model sustained a win rate exceeding 50% in AlpacaEval's performance evaluation, indicating that its general conversational ability and helpfulness remain intact despite prioritizing factual accuracy.
- My Thought
    - The compound reward is a multi-dimensional reward, so it can still have the inherent drawbacks of GRPO: it inherently *compresses* the reward signal, causing loss of information in the advantage estimates.
    - To increase robustness, the method of processing each sub-reward mentioned here can be used.
- Work Cited
    - Paper
    - Chat History